

Open Lounge Phone - schematic review (single board, variant lounge)

One 180 x 88 mm 4-layer board (owner decision 2026-09-27). Pages 2+: one sheet per subsystem; ICs as boxes with net labels, then the passives of that subsystem.

Subsystems

2. Power in: USB-C sink, protection, charger, 3V3 buck, 3V0 LDO, battery, supercap (94 parts)
3. MCU: ESP32-S3-WROOM-1U, reset/boot, USB-UART (CH340C), LED-data buffer, secure element (41 parts)
4. Audio: ES8311 DAC, ES7210 ADC, NS4150B amp, base mic, mic bias + privacy sense (51 parts)
5. Handset port: USB-C host (UAC), VBUS switch, ESD (8 parts)
6. Sensors and radar: hall hook, IR hook option, accelerometer, LD2410C (12 parts)
7. Side controls: VOL-, VOL+, MUTE (+ ESD) (10 parts)
8. Keys and LEDs: AW9523B, 12 hot-swap keys, 13 SK6812MINI-E, LED power gate (45 parts)
9. E-ink strip: 24-pin FPC + SSD1680 boost (DNP on display = none) (16 parts)
10. NFC tag, ambient light, privacy LED, Qwiic port (7 parts)

Rails

USB-C VBUS_C -> PTC F1 -> VBUS (TVS D3) -> BQ24074 U2 -> VSYS 4.4 V (battery B-option on VBAT, supercap on SCAP for Lounge)

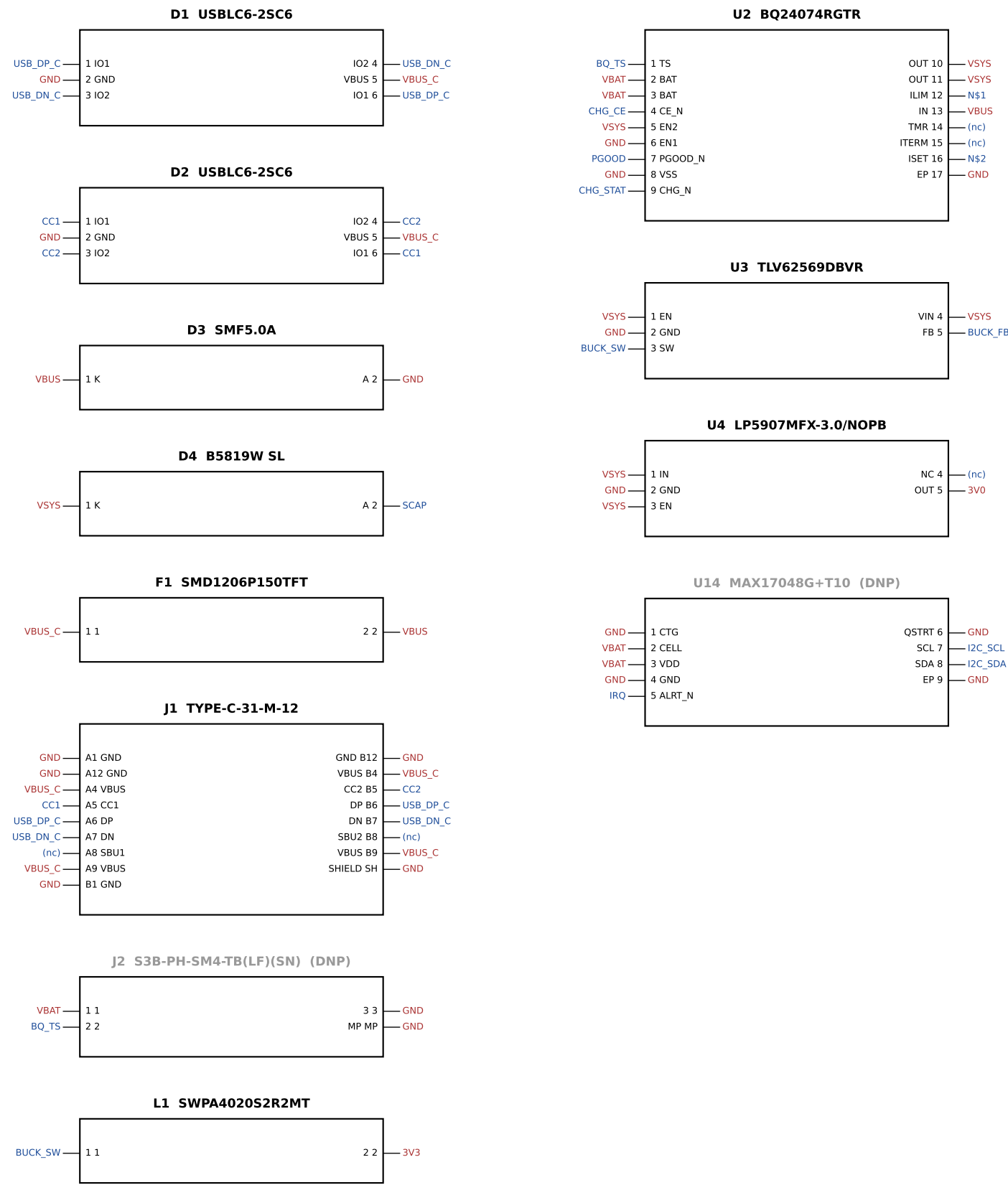
VSYS -> TLV62569 buck U3 -> 3V3 (digital, ESP32, sensors) ; VSYS -> LP5907 U4 -> 3V0 (codecs, mic bias)

VSYS -> P-FET gate -> VLED (13 x SK6812MINI-E) ; VBUS/VSYS diode-OR -> HS_VIN -> SY6280 U15 -> HS_VBUS (handset)

ESP32-S3 GPIO map (pin_table.yaml)

GPI00	B00T	10k pull-up + B00T button; runtime long-press = factory reset
GPI01	CC1_SENSE	
GPI02	CC2_SENSE	
GPI03	HS_VBUS_EN	handset port VBUS switch; strap only if EFUSE_STRAP_JTAG_SEL burned; 100k pull-down (
GPI04	HS_VBUS_SENSE	handset VBUS via 100k/100k
GPI05	H00K	
GPI06	LD_OUT	
GPI07	IRQ	
GPI08	I2C_SDA	
GPI09	I2C_SCL	
GPI010	EPD_CS	
GPI011	EPD_MOSI	
GPI012	EPD_SCK	
GPI013	EPD_DC	
GPI014	EPD_RST	
GPI015	EPD_BUSY	
GPI016	I2S_MCLK	
GPI017	I2S_BCLK	
GPI018	I2S_WS	
GPI019	HS_USB_DN	native USB OTG = host for the USB-C (UAC) handset port
GPI020	HS_USB_DP	
GPI021	I2S_DOUT	
GPI035	(unconnected)	reserved: octal PSRAM
GPI036	(unconnected)	reserved: octal PSRAM
GPI037	(unconnected)	reserved: octal PSRAM
GPI038	I2S_DIN	
GPI039	LD_RX	
GPI040	LD_TX	
GPI041	PA_EN	100k pull-down: amp off at boot
GPI042	LED_DATA	via SN74LV1T125 buffer on main
GPI043	U0TXD	
GPI044	U0RXD	
GPI045	CHG_CE	must be low at reset (3.3 V flash)
GPI046	LD_PWR_EN	
GPI047	CHG_STAT	
GPI048	PG00D	

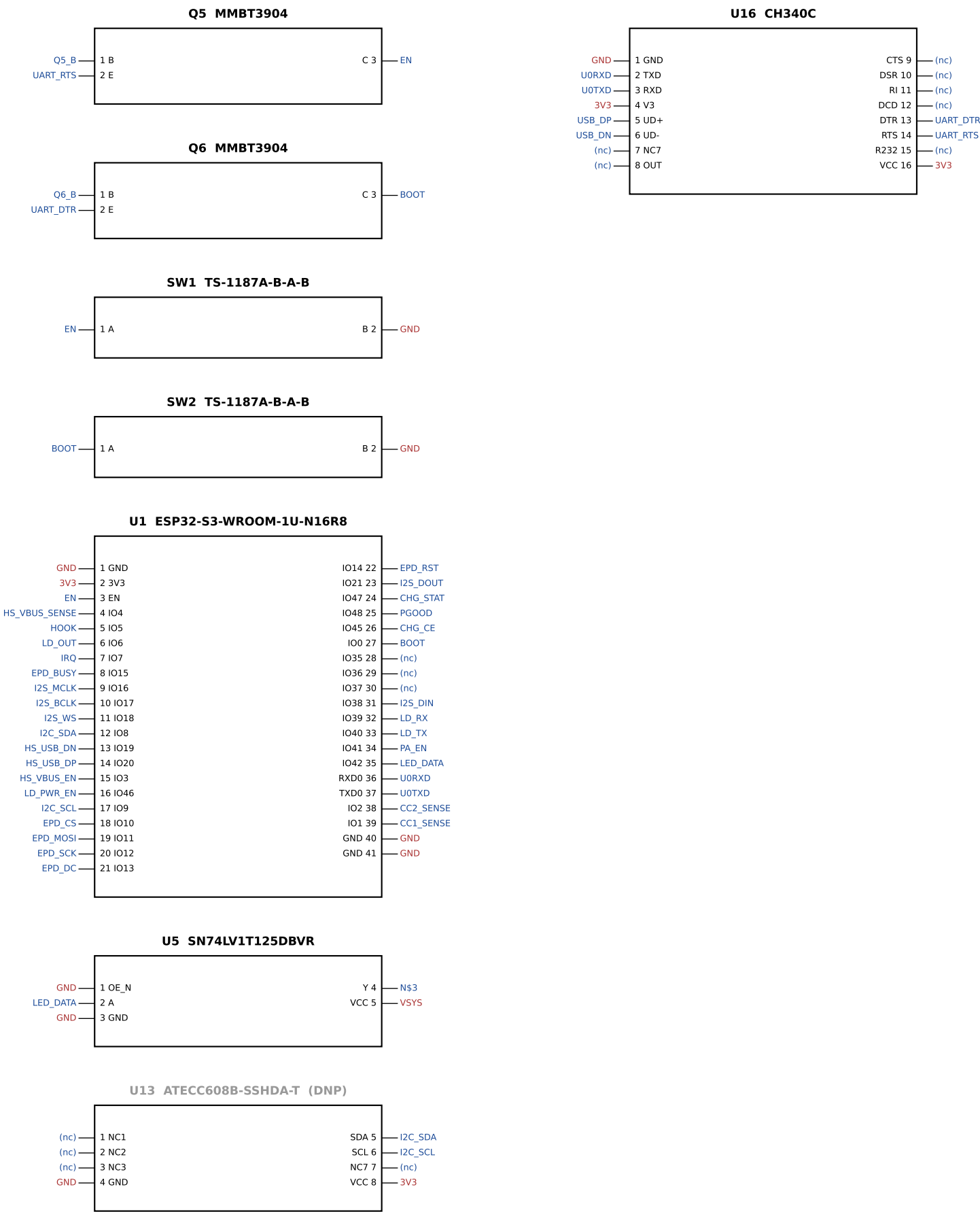
2. Power in: USB-C sink, protection, charger, 3V3 buck, 3V0 LDO, battery, supercap



2. Power in: USB-C sink, protection, charger, 3V3 buck, 3V0 LDO, battery, supercap: passives and test points

C1	10uF 10V X5R	VBUS - GND			R7	1.1k	N\$1 - GND
C2	1uF 50V X5R	VBUS - GND			R8	1.8k	N\$2 - GND
C3	10uF 10V X5R	VSYS - GND			R9	10k	BQ_TS - GND
C4	10uF 10V X5R	VSYS - GND			R10	10k	PGOOD - 3V3
C5	100nF 50V X7R	VSYS - GND			R11	10k	CHG_STAT - 3V3
C6	4.7uF 16V X5R	VBAT - GND			R12	10k	CHG_CE - GND
C7	SE-5R5-D474VYV	SCAP - GND			R13	47	VSYS - SCAP
C8	10uF 10V X5R	VSYS - GND			R14	100k	3V3 - BUCK_FB
C9	22uF 25V X5R	3V3 - GND			R15	22k	BUCK_FB - GND
C10	22uF 25V X5R	3V3 - GND			R22	4.7k	I2C_SDA - 3V3
C11	1uF 50V X5R	VSYS - GND			R23	4.7k	I2C_SCL - 3V3
C12	1uF 50V X5R	3V0 - GND			R24	10k	IRQ - 3V3
C13	10uF 10V X5R	3V0 - GND			TP2	VBUS	VBUS
C14	22uF 25V X5R	3V3 - GND			TP3	VSYS	VSYS
C15	100nF 50V X7R	3V3 - GND			TP4	3V3	3V3
C17	100nF 50V X7R	VSYS - GND			TP5	3V0	3V0
C18	100nF 50V X7R	3V3 - GND			TP6	GND	GND
C19	100nF 50V X7R	3V3 - GND			TP7	GND	GND
C20	1uF 50V X5R	3V0 - GND			TP8	GND	GND
C21	100nF 50V X7R	3V0 - GND			TP9	GND	GND
C25	100nF 50V X7R	3V3 - GND			TP20	I2C_SDA	I2C_SDA
C26	100nF 50V X7R	3V3 - GND			TP21	I2C_SCL	I2C_SCL
C27	1uF 50V X5R	3V0 - GND			TP26	HS_VBUS	HS_VBUS
C28	1uF 50V X5R	3V0 - GND					
C53	100uF 6.3V X5R	VSYS - GND					
C54	10uF 10V X5R	VSYS - GND					
C55	100nF 50V X7R	VSYS - GND					
C58	100nF 50V X7R	3V3 - GND					
C59	100nF 50V X7R	3V3 - GND					
C60	10uF 10V X5R	3V3 - GND					
C61	100nF 50V X7R	3V3 - GND	DNP				
C62	100nF 50V X7R	VBAT - GND	DNP				
C64	1uF 50V X5R	HS_VIN - GND					
C65	10uF 10V X5R	HS_VBUS - GND					
C67	100nF 50V X7R	3V3 - GND					
C69	100nF 50V X7R	3V3 - GND					
C70	22uF 25V X5R	VLED - GND					
C71	22uF 25V X5R	VLED - GND					
C72	100nF 50V X7R	VLED - GND					
C73	100nF 50V X7R	VLED - GND					
C74	100nF 50V X7R	VLED - GND					
C75	100nF 50V X7R	VLED - GND					
C76	100nF 50V X7R	VLED - GND					
C77	100nF 50V X7R	VLED - GND					
C78	100nF 50V X7R	VLED - GND					
C79	100nF 50V X7R	VLED - GND					
C80	100nF 50V X7R	VLED - GND					
C81	100nF 50V X7R	VLED - GND					
C82	100nF 50V X7R	VLED - GND					
C83	100nF 50V X7R	VLED - GND					
C84	100nF 50V X7R	VLED - GND					
C85	1uF 50V X5R	3V3 - GND					
C86	100nF 50V X7R	3V3 - GND					
C88	1uF 50V X5R	3V3 - GND					
C96	4.7uF 25V X5R	3V3 - GND					
R1	5.1k	CC1 - GND					
R2	5.1k	CC2 - GND					
R3	1k	CC1 - CC1_SENSE					
R4	1k	CC2 - CC2_SENSE					

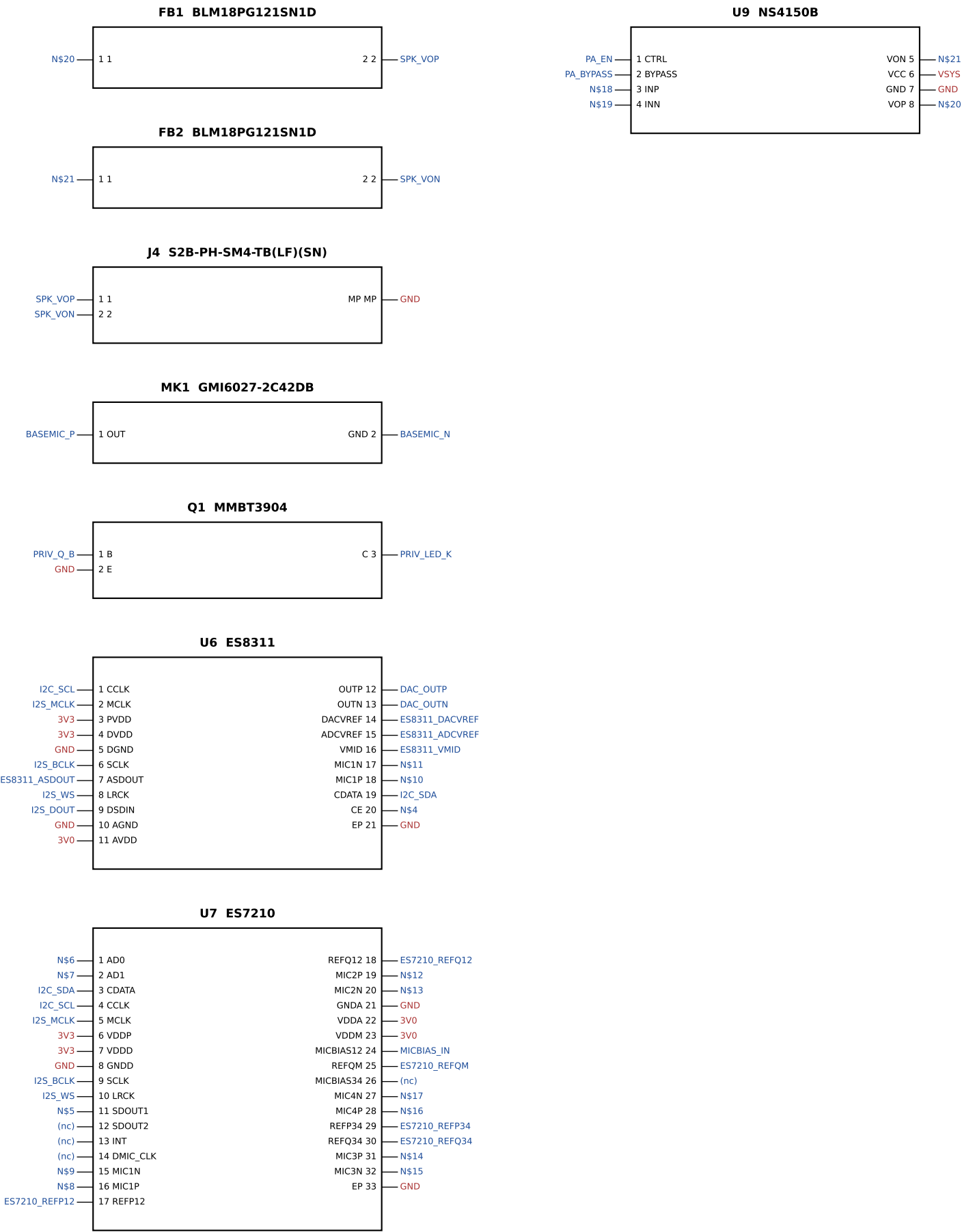
3. MCU: ESP32-S3-WROOM-1U, reset/boot, USB-UART (CH340C), LED-data buffer, secure element



3. MCU: ESP32-S3-WROOM-1U, reset/boot, USB-UART (CH340C), LED-data buffer, secure element: passives and test points

C16	1uF 50V X5R	EN - GND	
C66	100nF 50V X7R	HS_VBUS_SENSE - GND	
R5	0	USB_DP_C - USB_DP	
R6	0	USB_DN_C - USB_DN	
R16	10k	EN - 3V3	
R17	10k	BOOT - 3V3	
R18	100k	HS_VBUS_EN - GND	
R19	100k	PA_EN - GND	
R20	100k	LD_PWR_EN - GND	
R21	100k	LD_OUT - GND	
R25	100k	LED_DATA - GND	
R28	47	N\$5 - I2S_DIN	
R41	0	H00K_HALL - H00K	
R44	0	IR_PT_C - H00K	DNP
R47	1k	LD_TX - N\$22	
R48	1k	N\$23 - LD_RX	
R49	1k	N\$24 - LD_OUT	
R53	100k	HS_VBUS - HS_VBUS_SENSE	
R54	100k	HS_VBUS_SENSE - GND	
R55	10k	UART_DTR - Q5_B	
R56	10k	UART_RTS - Q6_B	
TP10	USB_DP	USB_DP	
TP11	USB_DN	USB_DN	
TP12	U0TXD	U0TXD	
TP13	U0RXD	U0RXD	
TP14	EN	EN	
TP15	BOOT	BOOT	
TP16	I2S_BCLK	I2S_BCLK	
TP17	I2S_WS	I2S_WS	
TP18	I2S_DIN	I2S_DIN	
TP19	I2S_DOUT	I2S_DOUT	
TP22	H00K	H00K	
TP23	PA_EN	PA_EN	

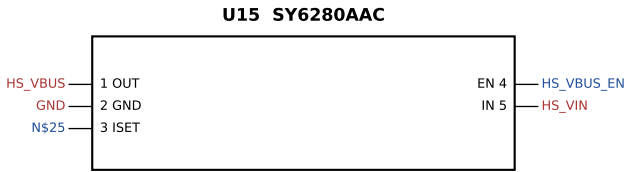
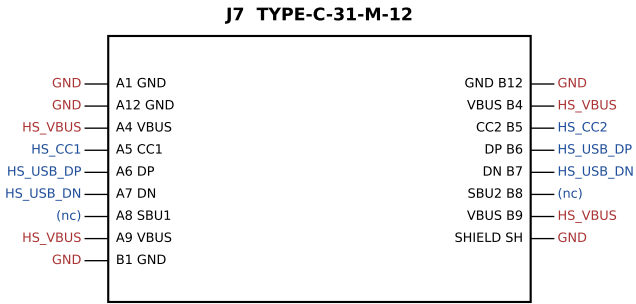
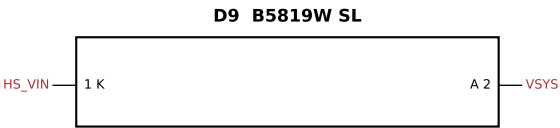
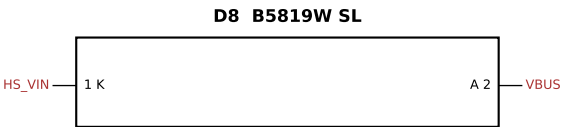
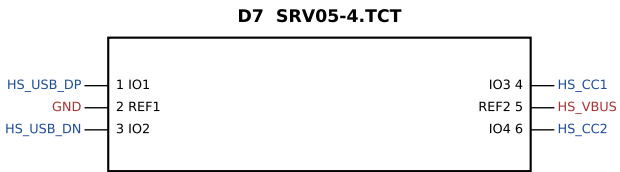
4. Audio: ES8311 DAC, ES7210 ADC, NS4150B amp, base mic, mic bias + privacy sense



4. Audio: ES8311 DAC, ES7210 ADC, NS4150B amp, base mic, mic bias + privacy sense: passives and test points

C22	1uF 50V X5R	ES8311_VMID - GND
C23	1uF 50V X5R	ES8311_ADCVREF - GND
C24	1uF 50V X5R	ES8311_DACVREF - GND
C29	1uF 50V X5R	ES7210_REFP12 - GND
C30	1uF 50V X5R	ES7210_REFQ12 - GND
C31	1uF 50V X5R	ES7210_REFP34 - GND
C32	1uF 50V X5R	ES7210_REFQ34 - GND
C33	1uF 50V X5R	ES7210_REFQM - GND
C34	1uF 50V X5R	MICBIAS_IN - GND
C35	10uF 10V X5R	MICBIAS_F - GND
C36	1uF 50V X5R	N\$8 - GND
C37	1uF 50V X5R	N\$9 - GND
C38	1uF 50V X5R	N\$10 - GND
C39	1uF 50V X5R	N\$11 - GND
C40	33pF 50V C0G	BASEMIC_P - GND
C41	1uF 50V X5R	BASEMIC_P - N\$12
C42	1uF 50V X5R	BASEMIC_N - N\$13
C43	470nF 25V X7R	DAC_OUTP - REF_AC_MIC3P
C44	1uF 50V X5R	REF_DIV_P - N\$14
C45	470nF 25V X7R	DAC_OUTN - REF_AC_MIC3N
C46	1uF 50V X5R	REF_DIV_N - N\$15
C47	100pF 50V C0G	REF_DIV_P - REF_DIV_N
C48	1uF 50V X5R	N\$16 - GND
C49	1uF 50V X5R	N\$17 - GND
C50	1uF 50V X5R	PA_BYPASS - GND
C51	100nF 50V X7R	DAC_OUTP - PA_IN_INP
C52	100nF 50V X7R	DAC_OUTN - PA_IN_INN
C56	220pF 50V X7R	SPK_VOP - GND
C57	220pF 50V X7R	SPK_VON - GND
NT1	NetTie	BASEMIC_N - GND
R27	10k	N\$4 - GND
R29	10k	N\$6 - GND
R30	10k	N\$7 - GND
R34	100k	PRIV_Q_B - GND
R35	2.2k	MICBIAS_F - BASEMIC_P
R36	20k	REF_AC_MIC3P - REF_DIV_P
R37	20k	REF_AC_MIC3N - REF_DIV_N
R38	4.3k	REF_DIV_P - REF_DIV_N
R39	150k	PA_IN_INP - N\$18
R40	150k	PA_IN_INN - N\$19
TP1	ES8311_ASDOUT	ES8311_ASDOUT
TP24	SPK_VOP	SPK_VOP
TP25	SPK_VON	SPK_VON

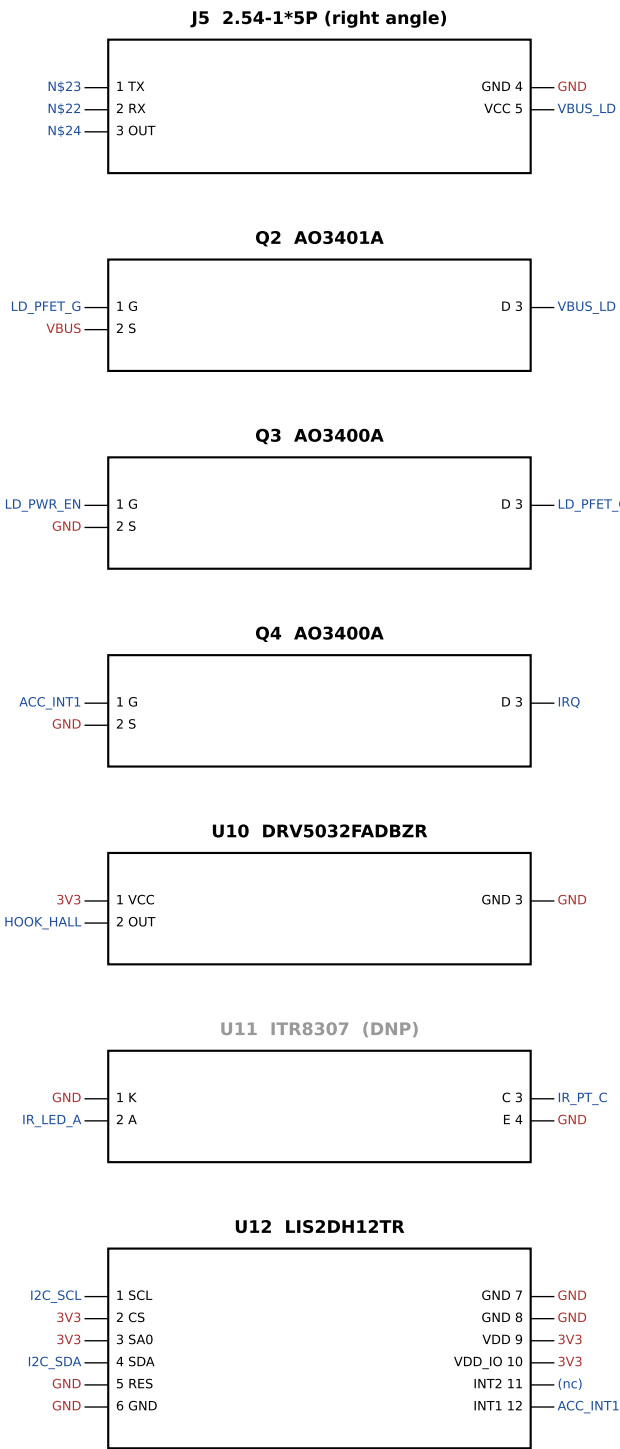
5. Handset port: USB-C host (UAC), VBUS switch, ESD



5. Handset port: USB-C host (UAC), VBUS switch, ESD: passives and test points

R50	33k	HS_CC1 - 3V3
R51	33k	HS_CC2 - 3V3
R52	15k	N\$25 - GND

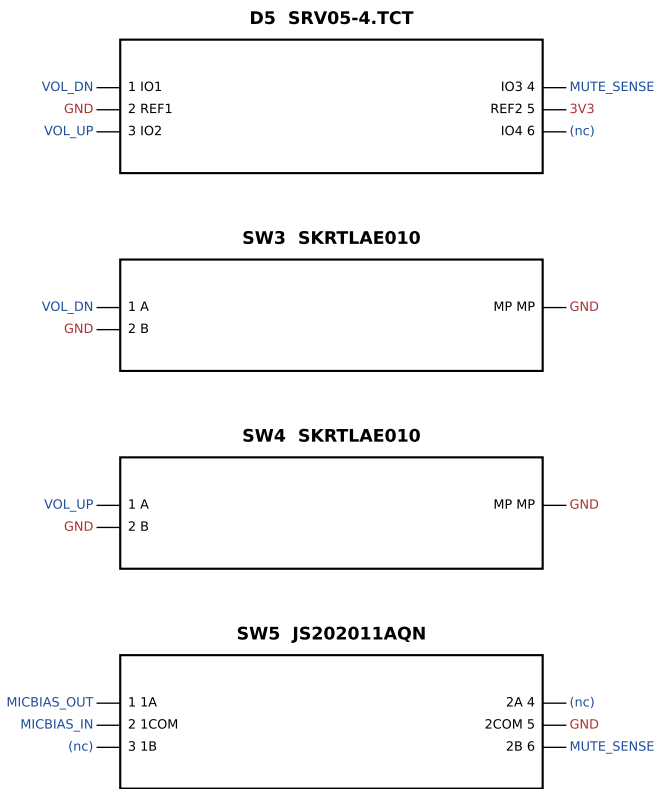
6. Sensors and radar: hall hook, IR hook option, accelerometer, LD2410C



6. Sensors and radar: hall hook, IR hook option, accelerometer, LD2410C: passives and test points

C63	10uF 10V X5R	VBUS_LD - GND	
R42	330	3V3 - IR_LED_A	DNP
R43	10k	IR_PT_C - 3V3	DNP
R45	100k	ACC_INT1 - GND	
R46	100k	VBUS - LD_PFET_G	

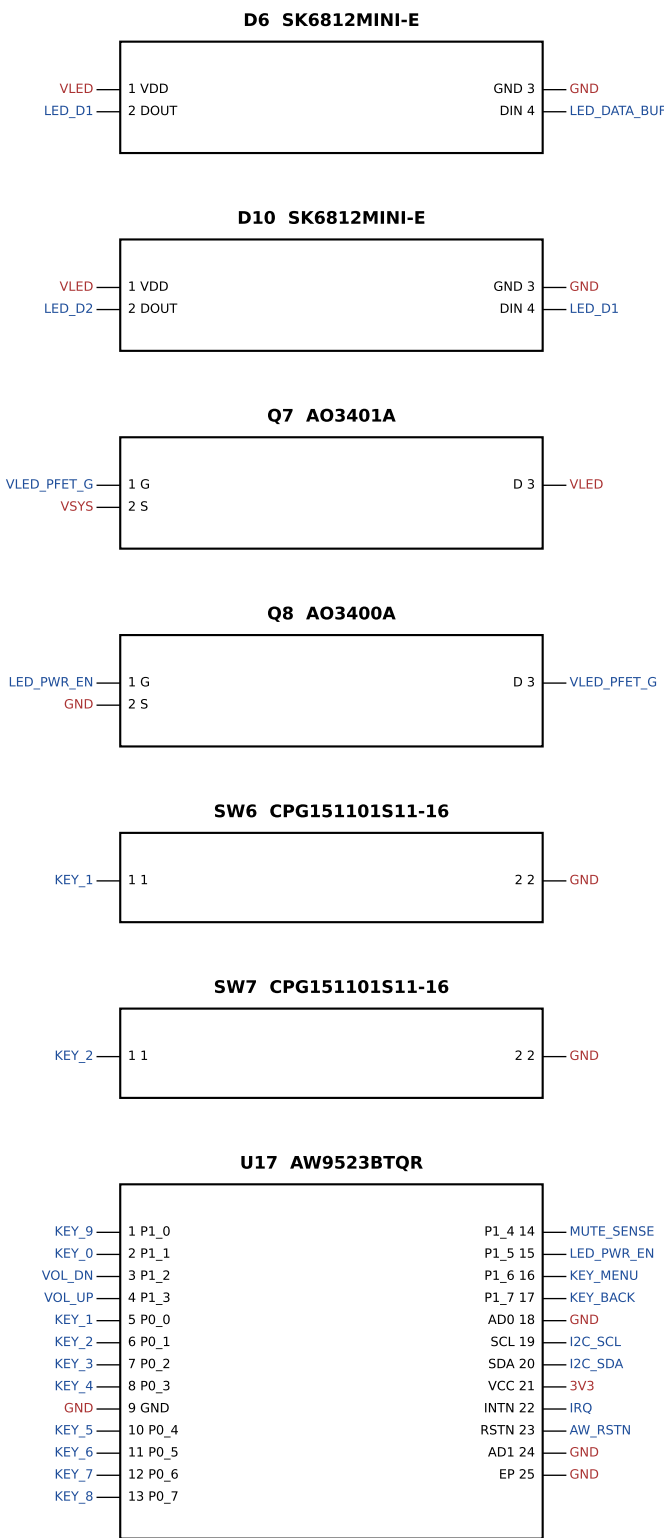
7. Side controls: VOL-, VOL+, MUTE (+ ESD)



7. Side controls: VOL-, VOL+, MUTE (+ ESD): passives and test points

R31	100k	MICBIAS_OUT - GND
R32	100	MICBIAS_OUT - MICBIAS_F
R33	22k	MICBIAS_OUT - PRIV_Q_B
R57	10k	VOL_DN - 3V3
R58	10k	VOL_UP - 3V3
R59	10k	MUTE_SENSE - 3V3

8. Keys and LEDs: AW9523B, 12 hot-swap keys, 13 SK6812MINI-E, LED power gate



13 x SK6812MINI-E in total (2 drawn): see the passives/refs list; 12 x HOTSWAP in total (2 drawn): see the passives/refs list

8. Keys and LEDs: AW9523B, 12 hot-swap keys, 13 SK6812MINI-E, LED power gate: every repeated part

D6: 1=VLED, 2=LED_D1, 3=GND, 4=LED_DATA_BUF

D10: 1=VLED, 2=LED_D2, 3=GND, 4=LED_D1

D11: 1=VLED, 2=LED_D3, 3=GND, 4=LED_D2

D12: 1=VLED, 2=LED_D4, 3=GND, 4=LED_D3

D13: 1=VLED, 2=LED_D5, 3=GND, 4=LED_D4

D14: 1=VLED, 2=LED_D6, 3=GND, 4=LED_D5

D15: 1=VLED, 2=LED_D7, 3=GND, 4=LED_D6

D16: 1=VLED, 2=LED_D8, 3=GND, 4=LED_D7

D17: 1=VLED, 2=LED_D9, 3=GND, 4=LED_D8

D18: 1=VLED, 2=LED_D10, 3=GND, 4=LED_D9

D19: 1=VLED, 2=LED_D11, 3=GND, 4=LED_D10

D20: 1=VLED, 2=LED_D12, 3=GND, 4=LED_D11

D21: 1=VLED, 2=nc, 3=GND, 4=LED_D12

SW6: 1=KEY_1, 2=GND

SW7: 1=KEY_2, 2=GND

SW8: 1=KEY_3, 2=GND

SW9: 1=KEY_4, 2=GND

SW10: 1=KEY_5, 2=GND

SW11: 1=KEY_6, 2=GND

SW12: 1=KEY_7, 2=GND

SW13: 1=KEY_8, 2=GND

SW14: 1=KEY_9, 2=GND

SW15: 1=KEY_0, 2=GND

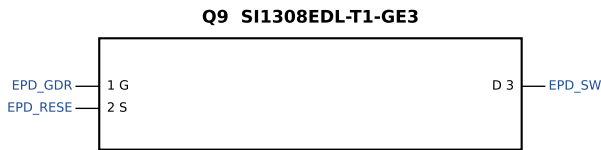
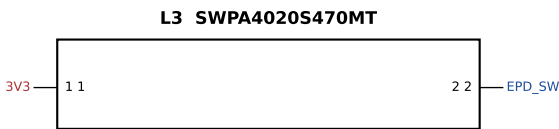
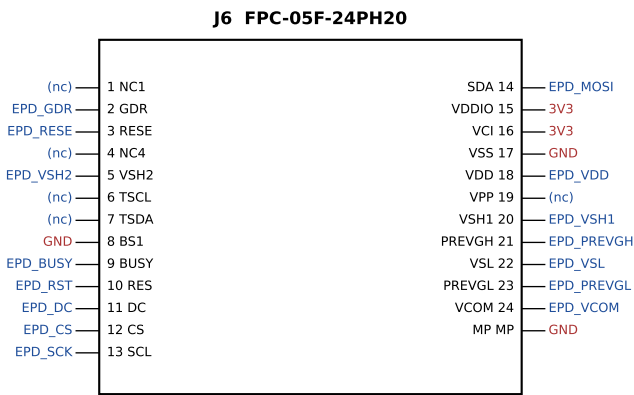
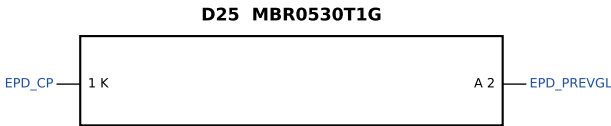
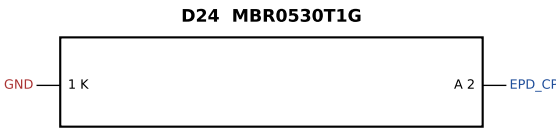
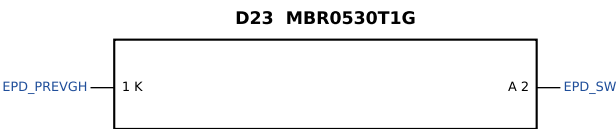
SW16: 1=KEY_MENU, 2=GND

SW17: 1=KEY_BACK, 2=GND

8. Keys and LEDs: AW9523B, 12 hot-swap keys, 13 SK6812MINI-E, LED power gate: passives and test points

C68	100nF 50V X7R	AW_RSTN - GND
R26	330	N\$3 - LED_DATA_BUF
R60	10k	AW_RSTN - 3V3
R61	10k	KEY_1 - 3V3
R62	10k	KEY_2 - 3V3
R63	10k	KEY_3 - 3V3
R64	10k	KEY_4 - 3V3
R65	10k	KEY_5 - 3V3
R66	10k	KEY_6 - 3V3
R67	10k	KEY_7 - 3V3
R68	10k	KEY_8 - 3V3
R69	10k	KEY_9 - 3V3
R70	10k	KEY_0 - 3V3
R71	10k	KEY_MENU - 3V3
R72	10k	KEY_BACK - 3V3
R73	100k	LED_PWR_EN - GND
R74	100k	VSYS - VLED_PFET_G

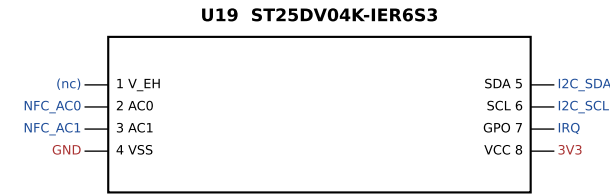
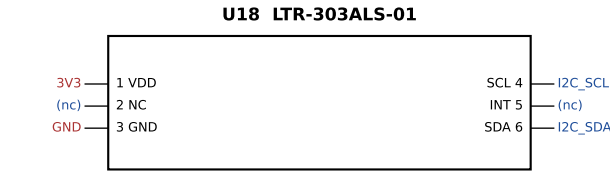
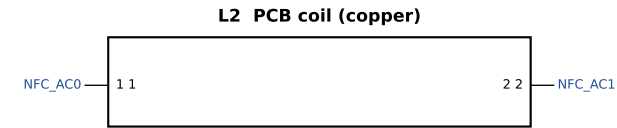
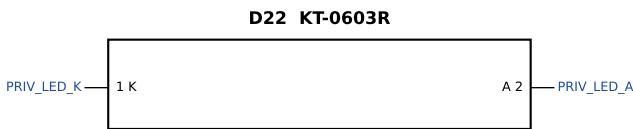
9. E-ink strip: 24-pin FPC + SSD1680 boost (DNP on display = none)



9. E-ink strip: 24-pin FPC + SSD1680 boost (DNP on display = none): passives and test points

C89	1uF 50V X5R	EPD_VSH2 - GND
C90	1uF 50V X5R	EPD_VDD - GND
C91	1uF 50V X5R	EPD_VSH1 - GND
C92	1uF 50V X5R	EPD_VSL - GND
C93	1uF 50V X5R	EPD_PREVGL - GND
C94	1uF 50V X5R	EPD_VCOM - GND
C95	1uF 50V X5R	EPD_PREVGH - GND
C97	4.7uF 25V X5R	EPD_SW - EPD_CP
R76	1M	EPD_GDR - GND
R77	2.2	EPD_RESE - GND

10. NFC tag, ambient light, privacy LED, Qwiic port



10. NFC tag, ambient light, privacy LED, Qwiic port: passives and test points

C87	22pF 50V C0G	NFC_AC0 - NFC_AC1	DNP
R75	470	3V3 - PRIV_LED_A	